APPM 4600 — HOMEWORK # 6

1. Consider the task of trying to find the solution (x, y) to the following nonlinear system of equations using an iterative method with a stopping tolerance of 10^{-4}.

$$\begin{aligned} 3x^2 + 4y^2 &= 1 \\ y^3 - 8x^3 &= 1 \end{aligned}$$

We use our iterative solvers with the following initial guesses

(i) $(x_0, y_0) = (-0.5, 0.25)$
(ii) $(x_0, y_0) = (-1.5, 1.25)$
(iii) $(x_0, y_0) = (-3, 3)$

(a) First use fixed point with the vector function

$$\mathsf{G}(\boldsymbol{x}) = \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 0.016 & -0.17 \\ 0.52 & -0.26 \end{bmatrix} \begin{bmatrix} 3x^2 + 4y^2 - 1 \\ y^3 - 8x^2 - 1 \end{bmatrix}.$$

What happens with the iteration for each initial guess? If the iteration diverges, what do you suspect is the reason for the performance?

(b) Finally, use Newton's method to solve the nonlinear system of equations. What happens with the iteration? Is the performance better or worse than the fixed point options?

2. The nonlinear system

$$\begin{cases} f(x, y) = x^2 + y^2 - 4 = 0 \\ g(x, y) = e^x + y - 1 = 0 \end{cases}$$

has two real solutions. In the last homework you used Newton's method to solve this problem with following initial guesses:

(i) $x = 1,\ y = 1$
(ii) $x = 1,\ y = -1$
(iii) $x = 0,\ y = 0$

Use the two quasi-Newton methods with the different initial guesses.
Is the performance better or worse that of Newton's methods?

3. Consider the nonlinear system

$$\begin{aligned} x + \cos(xyz) - 1 &= 0, \\ (1 - x)^{1/4} + y + 0.05z^2 - 0.15z - 1 &= 0, \\ -x^2 - 0.1y^2 + 0.01y + z - 1 &= 0. \end{aligned}$$

Using your own codes test the following three techniques for approximating the solution to the nonlinear system to within 10^{-6}:

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- Newton's method
- Steepest descent method
- First Steepest descent method with a stopping tolerance of 5×10^{-2}. Use the result of this as the initial guess for Newton's method.

Using the same initial guess, which technique converges the fastest? Try to explain the performance.

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